EngageNet Draft

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1 Architecture

1.1 Input

Let $x = \{x_1, \ldots, x_M\}$ with $x_i \in \mathbb{R}^{C_i \times L}$ for all $i \in \{1, \ldots, M\}$, where C_i is the number of acquisition channels for the i-th modality and L represents the number of time steps of the input signal.

1.2 Output

In [1], y is a discrete value corresponding to a category, which is then converted into a one-hot encoding $p(y|x) \in \mathbb{R}^N$ (where N is the number of categories). In our case, since we are performing regression, $y \in [0, 1] \implies p(y|x) \in [0, 1]^N$, so our model outputs the predicted category probability $p(\hat{y}|x) \in [0, 1]^N$.

1.3 Inner Workings

We need to capture contextual information from physiological time series simultaneously, and a unidirectional approach cannot fully capture the complex temporal patterns. The multimodal BiMamba network consists of two modules: the intra-modal BiMamba module and the inter-modal BiMamba module. The former is designed to model each modality independently, while the latter effectively fuses multimodal features.

1.3.1 Intra-modal BiMamba Module

First, an initial encoder $h_i = \text{InitEncoder}_i(x_i) \in \mathbb{R}^{L' \times C'_i}$ for all $i \in \{1, \ldots, M\}$ is used to extract shallow features from a specific modality, where C'_i is the number of output channels and L' is the feature length of the output signal.

The transposition of dimensions from $C_i \times L$ to $L' \times C'_i$ physically aligns the data structure with the sequence-first requirement of autoregressive models. By enforcing the sequence length L' as the primary operational dimension, the subsequent state-space and 1D convolutional operators can process the temporal dimension across the channel feature space.

The BiMamba module operates as follows:

1. The output of the gating mechanism is $g_i = \sigma(W_i^g h_i + b_i^g)$, where h_i is the hidden state, W_i^g is the weight matrix, b_i^g is the bias, and $\sigma(x) = x \cdot \dfrac{1}{1 + e^{-x}}$ (SiLU). Intuitively, g_i highlights emotion-relevant information while suppressing noise and redundancy.
2. BiMamba models temporal dependencies from both forward and backward perspectives using state-space modeling, thereby capturing richer contextual dynamics:

$$h_i^{\rightarrow} = g_i \otimes \text{SSM}_{\rightarrow}(\sigma(\text{Conv1D}_{\rightarrow}(W_i^h h_i + b_i^h))) \in \mathbb{R}^{L' \times C'_i}$$

$$h_i^{\leftarrow} = g_i \otimes \mathrm{SSM}_{\leftarrow}(\sigma(\mathrm{Conv1D}_{\leftarrow}(\mathrm{Rev}_t(W_i^h h_i + b_i^h)))) \in \mathbb{R}^{L' \times C_i'}$$

where $h_i^{\rightarrow}$ and $h_i^{\leftarrow}$ are the feature representations of the i-th modality in forward and backward modeling. $\mathrm{Rev}_t(\cdot)$ denotes flipping along the time dimension L'.

The operator $\otimes$ denotes the Hadamard product (element-wise multiplication), which explains why the dimensions remain strictly unchanged, unlike a direct sum ($\oplus$) or tensor product. W_i^h represents the weight matrix and b_i^h represents the bias.

$\mathrm{SSM}(\cdot)$ represents the Selective State Space Model. Formally, it maps a 1-dimensional sequence through a continuous linear time-invariant (LTI) system defined by the state equation $x'(t) = Ax(t) + Bu(t)$ and output $y(t) = Cx(t)$. To operate on discrete sequence steps, the continuous transition operator A and input operator B are discretized via a timescale parameter Δ, yielding the discrete operators $\bar{A} = \exp(\Delta A)$ and $\bar{B} = (\Delta A)^{-1}(\bar{A} - I)\Delta B$.

Unlike traditional LTI systems, Mamba introduces *selection* by making the discretized parameters $(\bar{B}, C, \Delta)$ functions of the input u_t. Specifically:

$$\Delta_t = \mathrm{softplus}(s_\Delta + \mathrm{Linear}(u_t))$$

$$B_t = \mathrm{Linear}(u_t), \quad C_t = \mathrm{Linear}(u_t)$$

where $\mathrm{softplus}(x) = \ln(1 + e^x)$ and $s_\Delta \in \mathbb{R}^D$ is a learned, input-independent bias term that defines the baseline timescale of the system. The inclusion of s_Δ within the *softplus* function ensures that the resulting Δ_t remains strictly positive, which is required for the stability of the discretization. This selection mechanism allows the discrete transition $x_t = \bar{A}_t x_{t-1} + \bar{B}_t u_t$ to dynamically adjust its temporal resolution - effectively allowing the model to "reset" its state or focus on transient patterns based on the specific content of the sequence u_t.

3. BiMamba applies a linear projection and a residual connection to integrate bidirectional features and stabilize training:

$$u_i = h_i + \left(W_i^o \left(\frac{h_i^{\rightarrow} + \mathrm{Rev}_t(h_i^{\leftarrow})}{2} \right) + b_i^o \right) \in \mathbb{R}^{L' \times C_i'}$$

which is the output of the intra-modal BiMamba module for the i-th modality. W_i^o is the weight matrix, and b_i^o is the bias.

1.3.2 Inter-modal BiMamba Module

Relying solely on intra-modal modeling neglects inter-modal correlations that are crucial for comprehensive multimodal understanding. Let the concatenation along the channel dimension be denoted by the operator $\circ$. We define the concatenated intermediate matrix U:

$$U = u_1 \circ u_2 \circ \cdots \circ u_M$$

where $U \in \mathbb{R}^{L' \times \sum_{i=1}^{M} C_i'}$. To enable state-space modeling across the modalities rather than the temporal space, we transpose this matrix:

$$m = U^\top = \mathrm{Transpose}(u_1 \circ u_2 \circ \cdots \circ u_M) \in \mathbb{R}^{(\sum C_i') \times L'}$$

Thus, each time step acts structurally as a "channel," while the concatenated modality dimension serves as the sequence length. We then input m into a BiMamba structure:

$$H = \mathrm{BiMamba}(m) \in \mathbb{R}^{\sum_{i=1}^{M} C_i' \times L'}$$

By performing bidirectional state modeling along the fused $\sum_{i=1}^{M} C_i'$ axis, the sequence transition operators effectively treat preceding modalities as the context basis for subsequent modalities.

The function $H = \text{BiMamba}(m)$ represents the application of the entire bidirectional Mamba block (All 1–3 steps of the intra-modal module above) to the fused matrix m. In this inter-modal stage, the "sequence length" is now the total number of concatenated channels $\sum C_i'$, and the "features" at each step are the temporal points L'. This allows the model to build a context where the state of one modality can directly influence the hidden state of another.

In the forward process, the hidden states built from modalities $\{1, \ldots, i-1\}$, while in the backward process the hidden states derived from the modalities $\{M, M-1, \ldots, i+1\}$ provide additional temporal information for the i-th modality.

2 Possible Improvement Ideas

1. Residual connection: we can replace the standard residual connection with DeepSeek AI's Manifold-Constrained Hyper-Connections (mHC) [2].

   Let the BiMamba transformation block be denoted as the layer function

   $$\mathcal{F}(h_i) = W_i^o \left(\frac{h_i^{\rightarrow} + \text{Rev}_t(h_i^{\leftarrow})}{2} \right) + b_i^o$$

   In the original architecture, the (residual) update is $u_i = h_i + \mathcal{F}(h_i)$.

   To apply mHC, we expand the residual stream width by a factor of n to form a multi-stream representation $\mathbf{h}_i \in \mathbb{R}^{n \times L' \times C_i'}$. The update rule becomes:

   $$u_i = \mathcal{H}_i^{res} \mathbf{h}_i + (\mathcal{H}_i^{post})^\top \mathcal{F}(\mathcal{H}_i^{pre} \mathbf{h}_i)$$

   where $\mathcal{H}_i^{pre} \in \mathbb{R}^{1 \times n}$ aggregates the multi-stream into a single input for $\mathcal{F}$, $\mathcal{H}_i^{post} \in \mathbb{R}^{1 \times n}$ maps the output back to the stream, and $\mathcal{H}_i^{res} \in \mathbb{R}^{n \times n}$ mixes features within the stream.

   The structural mappings $\mathcal{H}_i^{pre}$ and $\mathcal{H}_i^{post}$ are composed of dynamic, input-dependent linear projections and static, learnable biases. To prevent signal cancellation, they are mapped through a *sigmoid* activation σ:

   $$\mathcal{H}_i^{pre} = \sigma(\alpha_i^{pre}(\vec{\mathbf{h}}_i \varphi_i^{pre}) + b_i^{pre})$$

   $$\mathcal{H}_i^{post} = 2\sigma(\alpha_i^{post}(\vec{\mathbf{h}}_i \varphi_i^{post}) + b_i^{post})$$

   where $\vec{\mathbf{h}}_i$ is the flattened input tensor, φ represents the linear projection weights, and α acts as a learnable gating factor.

   To maintain signal conservation, $\mathcal{H}_i^{res}$ is projected onto the Birkhoff polytope using the Sinkhorn-Knopp algorithm. The Birkhoff polytope is the convex hull of all permutation matrices, consisting entirely of doubly stochastic matrices (square matrices with non-negative real entries where each row and column sums exactly to 1).

   The projection is achieved iteratively:

   $$\mathcal{H}_i^{res} = \text{Sinkhorn-Knopp}(\tilde{\mathcal{H}}_i^{res})$$

   Given an initial positive matrix $M^{(0)} = \exp(\tilde{\mathcal{H}}_i^{res})$, the Sinkhorn-Knopp algorithm alternately normalizes the rows and columns until convergence. Constraining $\mathcal{H}_i^{res}$ to the

Birkhoff polytope ensures its spectral norm is strictly bounded by 1 ($||\mathcal{H}_i^{res}||_2 \leq 1$). This property guarantees stability across deep layers, strictly regulating the vector norm and preventing the signal from exploding or vanishing within the Hilbert space of our feature representations.

3 Engagement Regression Adaptation

3.1 Differential Entropy and the Failure of Gaussian Assumptions

To adapt the entropy-based selection mechanism of BiMamba-TTA for continuous engagement regression, we transition from Shannon entropy to **differential entropy**. For a continuous random variable y with probability density $p(y)$, the differential entropy is defined as:

$$H(y) = -\int_{\mathcal{Y}} p(y) \ln p(y)\, dy$$

While a Gaussian assumption $p(y|x) = \mathcal{N}(\mu, \sigma^2)$ simplifies this to $H(y) = \frac{1}{2}\ln(2\pi e\sigma^2)$, it is fundamentally inappropriate for engagement modeling. Engagement scores are strictly bounded within the interval $[0, 1]$. A Gaussian distribution assigns non-zero probability mass to values in $(-\infty, 0)$ and $(1, \infty)$, leading to "probability leakage" and poor calibration near the boundaries.

3.2 Beta Regression Framework

To respect the support $\mathcal{Y} \in [0, 1]$, we model the predictive distribution using a **Beta distribution**:

$$p(y|\alpha, \beta) = \frac{y^{\alpha-1}(1-y)^{\beta-1}}{B(\alpha, \beta)}, \quad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

The network outputs these parameters via a softplus activation: $\alpha = \text{softplus}(z_\alpha) + 1$ and $\beta = \text{softplus}(z_\beta) + 1$. The $+1$ offset ensures the distribution remains unimodal ($\alpha, \beta > 1$).

The predictive mean $\hat{\mu}$ and aleatoric uncertainty $\text{Var}[y]$ are:

$$\hat{\mu} = \frac{\alpha}{\alpha+\beta}, \quad \text{Var}[y] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

The Beta variance is naturally boundary-aware, satisfying:

$$\lim_{\hat{\mu}\to 0} \text{Var}[y] = 0 \quad \text{and} \quad \lim_{\hat{\mu}\to 1} \text{Var}[y] = 0$$

This provides a principled measure of confidence that scales with the proximity to the interval limits.

3.3 Test-Time Adaptation (TTA) and Adaptive Thresholds

Following the differential-entropy principle of the original BiMamba-TTA paper [1], we replace Shannon entropy with a regression-compatible uncertainty proxy. For the Beta predictive distribution $p(y \mid x) = \text{Beta}(\hat{\alpha}(x), \hat{\beta}(x))$ that respects the support $y \in [0, 1]$, we define

$$U(x) := \text{Var}[y] = \frac{\hat{\alpha}(x)\hat{\beta}(x)}{(\hat{\alpha}(x)+\hat{\beta}(x))^2(\hat{\alpha}(x)+\hat{\beta}(x)+1)}.$$

3.3.1 Why Predictive Variance is a Strictly Monotonic Proxy for Differential Entropy

This choice is justified because variance is a *strictly monotonic* function of the differential entropy $H(p)$ of the Beta distribution. To see this explicitly, re-parameterise the Beta distribution by its mean and concentration:

$$\mu = \frac{\hat{\alpha}}{\hat{\alpha} + \hat{\beta}}, \qquad \kappa = \hat{\alpha} + \hat{\beta}.$$

Substituting these parameters yields the simple expression

$$U(x) = \frac{\mu(1-\mu)}{\kappa + 1}.$$

For any fixed mean $\mu \in (0, 1)$, the partial derivative

$$\frac{\partial U}{\partial \kappa} = -\frac{\mu(1-\mu)}{(\kappa+1)^2} < 0$$

shows that variance is strictly decreasing in κ: larger concentration κ produces a tighter distribution and therefore smaller variance.

The differential entropy of the Beta distribution has the closed-form expression

$$H(p) = \ln B(\hat{\alpha}, \hat{\beta}) - (\hat{\alpha} - 1)\psi(\hat{\alpha}) - (\hat{\beta} - 1)\psi(\hat{\beta}) + (\hat{\alpha} + \hat{\beta} - 2)\psi(\hat{\alpha} + \hat{\beta}),$$

where $B(\cdot, \cdot)$ is the Beta function and $\psi(\cdot)$ is the digamma function. Substituting the same re-parameterisation $\hat{\alpha} = \mu\kappa$, $\hat{\beta} = (1-\mu)\kappa$ and taking the derivative with respect to κ likewise yields

$$\frac{\partial H}{\partial \kappa} < 0$$

for any fixed $\mu \in (0, 1)$. Thus both $U(x)$ and $H(p)$ are strictly decreasing functions of κ: larger concentration (tighter distribution) produces both smaller variance *and* smaller differential entropy. Consequently, the ranking of samples by uncertainty is identical whether one uses the full (expensive) differential entropy expression or simply the predictive variance.

Variance is numerically stable because its formula consists only of basic arithmetic operations (no logarithms, no digamma or Gamma functions, and no risk of underflow when $\hat{\alpha}$ or $\hat{\beta}$ become large). It is boundary-aware: $\lim_{\mu \to 0^+} U = 0$ and $\lim_{\mu \to 1^-} U = 0$, which matches the intuition that a prediction that is certain to be near the extremes of engagement (fully disengaged or fully engaged) should carry low uncertainty. Finally, variance is already required for the Beta negative log-likelihood (Beta NLL) loss used during training:

$$\mathcal{L}_{\text{NLL}}(x, y) = -[(\hat{\alpha} - 1)\ln y + (\hat{\beta} - 1)\ln(1 - y) - \ln B(\hat{\alpha}, \hat{\beta})].$$

Because the variance appears directly in the predictive distribution (and therefore in every forward pass), it incurs no extra computation at inference time. All these properties make variance the natural, cheap uncertainty proxy for regression.

A sample is selected for the TTA backward pass if the multimodal prediction is confident (low uncertainty) but the unimodal branches are uncertain (high uncertainty), i.e.,

$$S(x) = \mathbb{I}\Big\{U_{\text{multi}}(x) \le \gamma_m(t) \text{ AND } \sum_{i=1}^{M} w_i U_i(x) \ge \gamma_u(t)\Big\}.$$

To handle non-stationary streaming data (i.e., test-time engagement signals that arrive continuously frame-by-frame and whose statistical distribution drifts over time due to fatigue, cultural differences, sensor drift, or changing speaker roles), the thresholds $\gamma(t)$ are updated adaptively using a moving window of the most recent K samples:

$$\gamma(t) = \text{Percentile}\big(\mathcal{U}_t,\, p\big), \quad \mathcal{U}_t = \big\{U(\tau) \mid \tau \in [t-K, t]\big\},$$

where p is the percentile rank (e.g., $p = 0.8$) that defines the "confidence" cutoff relative to the recent noise floor of the data stream. Here "non-stationary streaming data" means the incoming test distribution is not fixed but changes gradually; the moving window $\mathcal{U}_t$ tracks this drift in real time. The "percentile rank $p = 0.8$" selects the value below which 80 % of the recent uncertainties lie, so the threshold automatically adapts to the current level of noise. The term "confidence cutoff" means that samples with $U_{\text{multi}}(x)$ below this adaptive value are considered sufficiently reliable for adaptation. The "recent noise floor" refers to the lowest typical uncertainty observed in the immediate past window - using the percentile of this floor makes the threshold robust to gradual changes in signal quality.

3.4 Uncertainty Estimation Methods

Uncertainty is formally quantified as the variance of the predictive distribution. We utilize three complementary methods that together capture both aleatoric and epistemic uncertainty. All three are expressed in terms of the Beta predictive distribution $p(y \mid x) = \text{Beta}(\hat{\alpha}(x), \hat{\beta}(x))$.

1. **Parametric Beta (Aleatoric Uncertainty):** This is the direct predictive variance obtained from the dual-head Beta output:

$$U(x) = \text{Var}[y] = \frac{\hat{\alpha}(x)\hat{\beta}(x)}{(\hat{\alpha}(x) + \hat{\beta}(x))^2(\hat{\alpha}(x) + \hat{\beta}(x) + 1)}.$$

   Because the variance is computed analytically from the learned shape parameters, it is the fastest and default choice for the TTA filter.

2. **Monte Carlo Dropout (Epistemic Uncertainty):** We keep dropout active at inference and draw T stochastic forward passes (typically $T = 10$–20). For each pass t we obtain a Beta distribution with parameters $(\hat{\alpha}^{(t)}, \hat{\beta}^{(t)})$ and corresponding mean $\hat{\mu}^{(t)} = \dfrac{\hat{\alpha}^{(t)}}{\hat{\alpha}^{(t)} + \hat{\beta}^{(t)}}$. The epistemic uncertainty is the variance of these means:

$$U_{\text{epi}}(x) = \frac{1}{T}\sum_{t=1}^{T}\big(\hat{\mu}^{(t)}(x) - \bar{\mu}(x)\big)^2,$$

   where $\bar{\mu}(x) = \frac{1}{T}\sum_{t=1}^{T}\hat{\mu}^{(t)}(x)$. This term quantifies the model's "lack of knowledge" (how sensitive the prediction is to random weight masks) and is especially useful under distribution shift (NoXi $\rightarrow$ NoXi+J).

3. **Total Uncertainty (Hybrid):** Using the law of total variance, the full predictive uncertainty decomposes as

$$U_{\text{total}}(x) = \underbrace{\frac{1}{T}\sum_{t=1}^{T}\text{Var}\big[\text{Beta}(\hat{\alpha}^{(t)}, \hat{\beta}^{(t)})\big]}_{\text{aleatoric (data)}} + \underbrace{\frac{1}{T}\sum_{t=1}^{T}\big(\hat{\mu}^{(t)}(x) - \bar{\mu}(x)\big)^2}_{\text{epistemic (model)}}.$$

   The decomposition is mainly used as a diagnostic to monitor whether the shift is mostly epistemic.

Note that the symbol $U(x)$ in the TTA filter always refers to one of the three quantities above (most often the parametric Beta variance for speed or the hybrid total variance when epistemic information is needed). The law of total variance guarantees that U_{total} is the correct variance of the mixture distribution induced by MC dropout.

3.5 Test-Time Loss Derivations

The TTA objective enforces consistency between the unimodal student heads and the multimodal teacher while still supervising the students toward the multimodal prediction. We define the **Mutual Information Sharing (MIS) loss** as the KL divergence that pushes each unimodal Beta distribution toward the multimodal one:

$$\mathcal{L}_{\text{mis}} = \sum_{i=1}^{M} D_{KL}\big(\text{Beta}(\alpha_i, \beta_i) \parallel \text{Beta}(\alpha_{\text{multi}}, \beta_{\text{multi}})\big),$$

where the closed-form Beta KL divergence, with $B_i := B(\alpha_i, \beta_i)$, is:

$$D_{KL}(B_1 \parallel B_2) = \ln \frac{B(\alpha_2, \beta_2)}{B(\alpha_1, \beta_1)} + (\alpha_1 - \alpha_2)\psi(\alpha_1) + (\beta_1 - \beta_2)\psi(\beta_1) + (\alpha_2 - \alpha_1 + \beta_2 - \beta_1)\psi(\alpha_1 + \beta_1)$$

The final TTA loss combines the MIS consistency term with a supervision term that treats the multimodal mean as a pseudo-target for each unimodal head:

$$\mathcal{L}_{TTA} = \mathcal{L}_{\text{mis}} + \lambda \sum_{i=1}^{M} \text{NLL}_{\text{Beta}}\big(\hat{\mu}_{\text{multi}},\, \alpha_i, \beta_i\big).$$

Here $\lambda > 0$ is a hyperparameter that controls the relative importance of the two terms: larger λ emphasises direct supervision from the multimodal teacher, while smaller λ puts more weight on cross-modal consistency. The term $\text{NLL}_{\text{Beta}}(\hat{\mu}_{\text{multi}}, \alpha_i, \beta_i)$ is the Beta negative log-likelihood evaluated at the multimodal mean $\hat{\mu}_{\text{multi}}$ (treated as a pseudo-label) with the unimodal shape parameters.

3.6 Surgical Fine-Tuning and Variance Starvation

We only update specific surgical layers to prevent catastrophic forgetting:

- **BatchNorm layers:** These encode per-channel activation statistics, allowing for immediate adaptation to domain shifts.
- **First convolution in InitEncoder:** Re-calibrates low-level feature extraction (e.g., sensor noise) before temporal/modal mixing.
- **First FC layer in Inter-modal BiMamba:** Recalibrates cross-modal fusion (e.g., adjusting the weight of audio vs. visual cues for different cultures).

This surgical approach mitigates the **Variance Starvation Problem (VSP)**. In VSP, a model minimises the Beta NLL by artificially inflating $\text{Var}[y]$ (i.e., reducing the concentration $\kappa = \hat{\alpha} + \hat{\beta}$) on hard samples; the gradient to the mean head then vanishes because the loss term becomes insensitive to prediction error. By freezing the majority of the backbone and updating only the listed surgical layers, we force the model to improve genuine feature alignment rather than using uncertainty as a "slack variable" to bypass difficult data.

References

[1] Ziyu Jia et al., *A Multimodal BiMamba Network with Test-Time Adaptation for Emotion Recognition Based on Physiological Signals*, 39th Conference on Neural Information Processing Systems (NeurIPS 2025).

[2] Zhenda Xie et al., *mHC: Manifold-Constrained Hyper-Connections*, arXiv preprint arXiv:2512.24880v2 (2026).